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Role-AI/ML Intern

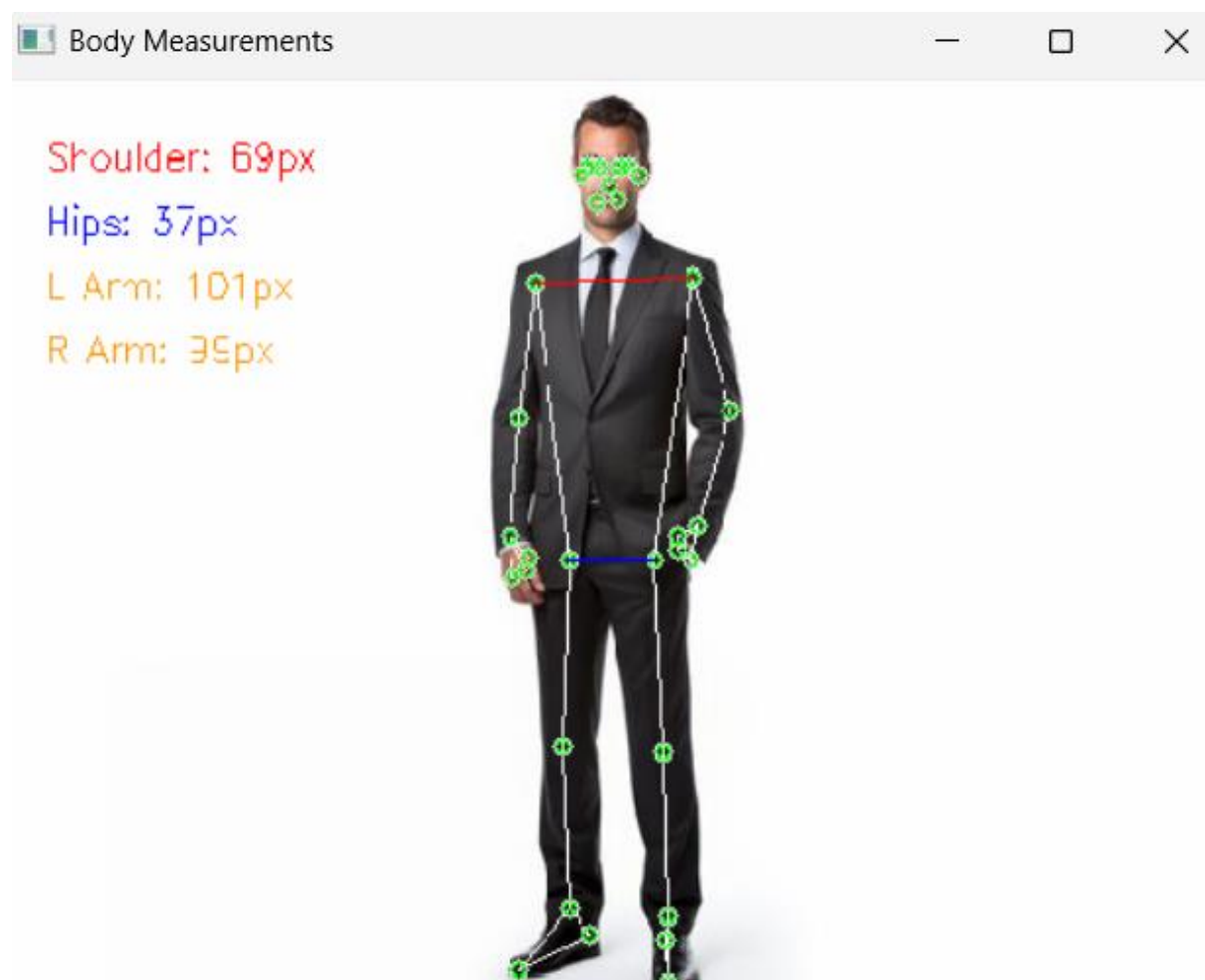
Day 7 – Project Module 1: Landmark Detection

For this task, I worked on detecting human body landmarks using MediaPipe and OpenCV. The system identifies important body points from an image or webcam input and connects them to create a body structure. This helps in understanding body posture and movement in a simple visual way.

Two approaches can be implemented:

- Webcam Detection – The system captures live camera input and detects body landmarks in real time. The landmarks move according to body movement and display the body structure instantly.
- Image-Based Detection – The system takes a saved image as input, detects body landmarks, and generates the output image with body points and skeletal connections.

OUTPUT

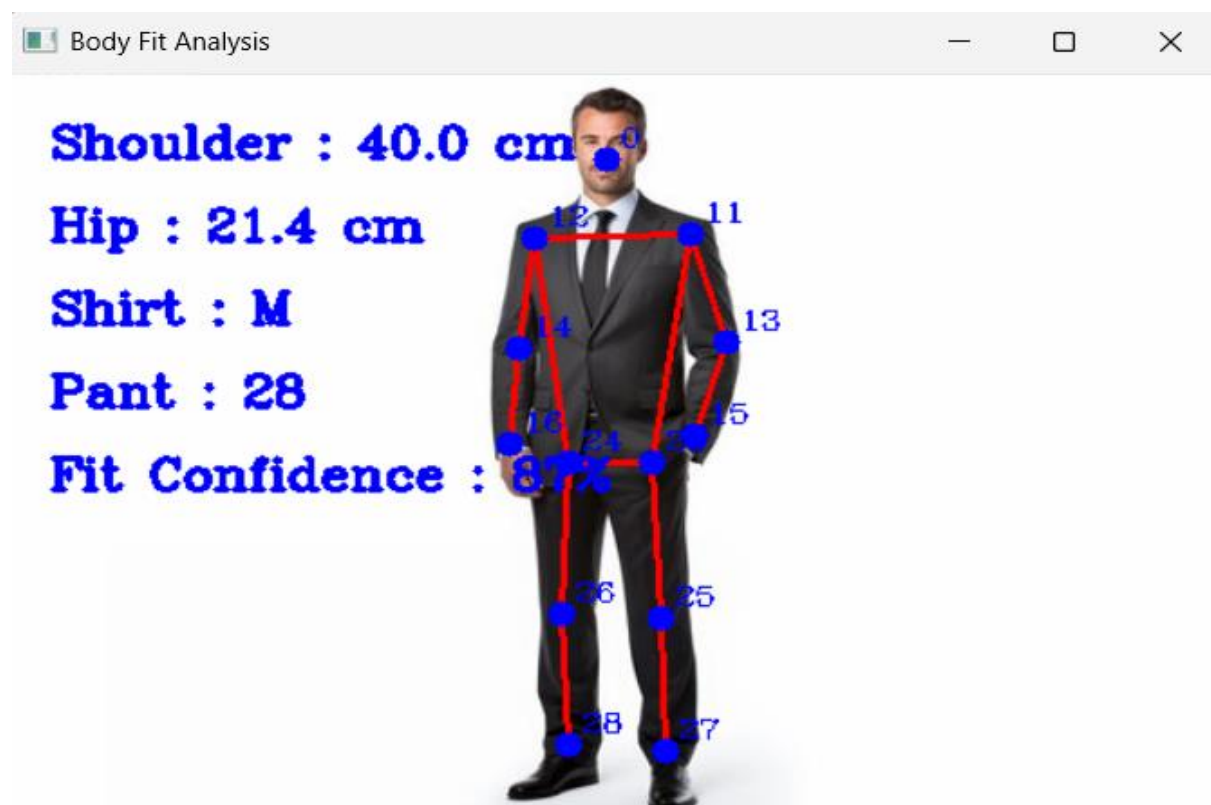


Size Recommendation System

For this task, I developed a Human Body Landmark Detection and Basic Body Measurement system using Python, OpenCV, and Media Pipe. The system captures body posture through webcam/image input and detects important body landmarks automatically. Based on detected points, basic body measurements and clothing size recommendations were generated along with fit confidence estimation and also calculated a Fit Confidence score, which indicates how confidently the recommended size matches the detected body proportions.

Sample Result:

- Shoulder Width: 40.0 cm
- Hip Width: 21.4 cm
- Recommended Shirt Size: M
- Recommended Pant Size: 28
- Fit Confidence: 87%



Module 2 – Angle Pose Detection

This module is developed to detect human body posture and calculate body joint angles using computer vision techniques. The system takes an input image and identifies important body landmarks such as shoulders, elbows, wrists, hips, knees, and ankles using Media Pipe Pose estimation.

After detecting landmarks, mathematical calculations are applied to determine the angle formed between three body points. For example, elbow angle is calculated using shoulder–elbow–wrist coordinates and knee angle is calculated using hip–knee–ankle coordinates.

The calculated angles help analyse body posture, movement, and alignment. This approach can be useful in applications such as:

- Fitness and exercise posture analysis
- Body measurement estimation
- Virtual try-on systems
- Health and rehabilitation monitoring
- Human pose tracking applications

How does Angle Pose Detection help in this project?

- Detects body posture automatically from images.
- Calculates angles between body joints accurately.
- Helps measure body parts like shoulders, hips, and arms.
- Reduces manual measurement effort.

Tools and Technologies Used:

- **Python** – Programming language
- **OpenCV** – Image processing
- **Media Pipe Pose** – Human pose detection
- **NumPy** – Coordinate calculations

